Md Niaz Morshed

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Summary

My research interests focus on software security, empirical software engineering, and AI-assisted secure software development. My work examines how peer code review practices influence the detection of security vulnerabilities in open-source software and develops data-driven and AI-based approaches to support reviewers in identifying security issues. I explore how generative AI and large language models can enhance reviewer reasoning and provide context-aware security insights during code review. My research bridges human expertise and automated analysis to improve the effectiveness of security-focused code review in real-world settings. I employ empirical research methods, including surveys, systematic literature reviews, mining software repositories, qualitative coding, and statistical analysis.

Education

2021 – present **Ph.D. Candidate in Computer Science, University of Alabama**
Advisor: Dr. Jeffrey Carver
Thesis title: *Empirical Investigation and LLM-Based Support for Security-Focused Code Review in Open-Source Software.*

2021 – 2025 **M.S. in Computer Science, University of Alabama**

2007 – 2010 **B.Sc in Computer Science and Engineering, Dhaka University of Engineering & Technology**

Working Experience

2023 – present **Graduate Research Assistant** at University of Alabama
I contributed to the Cooperative Institute for Research to Operations in Hydrology (CIROH) project. I designed and conducted surveys with CIROH researchers and developers to understand current water prediction practices and community needs. I analyzed survey responses to describe how research and operational teams support hydrologic forecasting. I am contributing to building training materials for the Next Generation Water Resources Modeling Framework (NextGen), which supports a more flexible approach to hydrologic modeling.

2021(June)–2022 **Graduate Research Assistant** at University of Alabama
I contributed to the INTERSECT project (INnovative Training Enabled by a Research Software Engineering Community of Trainers) by collecting, curating, and organizing research software engineering training resources, including slides, videos, and notebooks, and integrating them into the project website. I supported the design and development of workshop and bootcamp materials to improve the accessibility and effectiveness of research software engineering education.

2021 (Jan - May) **Graduate Teaching Assistant** at University of Miami
I served as a teaching assistant for Python course, responsible for evaluating assignments and supporting course delivery.

2015 – 2020 **Assistant Professor** at Dhaka University of Engineering & Technology, Bangladesh
I taught undergraduate courses including Algorithm Design & Analysis, Compiler Design, Simulation & Modeling, and Artificial Intelligence. I supervised undergraduate student thesis and projects. In addition to teaching, I served as Undergraduate Coordinator and contributed to academic and administrative activities, including curriculum design, Assistant Provost of student residence Hall, and moderator of computer society.

Working Experience (continued)

2012 – 2015 **Lecturer** at Dhaka University of Engineering & Technology, Bangladesh
I taught undergraduate courses including Discrete Mathematics, Data Structure, Numerical Methods, Microprocessor & Assembly Language, Digital System Design, and Neural Network & Fuzzy system. In addition to teaching, I served as member of undergraduate committee and took on several student affairs roles including advisor to the robotics, debate and chess clubs.

2013 – 2020 **Guest Lecturer** at Bangladesh Open University, Bangladesh
I taught undergraduate courses including Discrete Mathematics, Data Structure, Compiler Design, Microprocessor & Assembly Language and Artificial Intelligence.

2011 – 2012 **Senior Officer (IT)** at Bangladesh Development Bank Limited, Bangladesh
I Managed and supported IT operations, including system maintenance, software development, and database management.

Research Publications

Journal Articles

1. M. N. Morshed and J. C. Carver, "Use of peer code review to identify potential security vulnerabilities in open source software: A systematic literature review," *Information and Software Technology*, vol. 199, p. 108 256, 2026, ISSN: 0950-5849. DOI: https://doi.org/10.1016/j.infsof.2026.108256.
2. I. Shohidul, M. Niaz, S. Islam, and A. Mejbahul, "An experimental analysis of random early discard (red) queue for congestion control," *International Journal of Computer Applications*, vol. 15, Feb. 2011. DOI: 10.5120/1920-2563.

In Progress

1. M. Md Niaz and J. C. Carver, *Understanding information needs when looking for security issues during code review in open-source software (oss): A survey*, 2026.

Skills

Research Methods	Empirical software engineering, systematic literature review, survey design and analysis, open coding, thematic analysis, mixed-methods research, grounded theory approach, statistical analysis, and mining software repositories.
Natural Language Processing	Large language model, Transformers, fine-tuning, text classification.
Machine Learning	Classical learning, deep learning.
Cybersecurity	Vulnerability analysis, MITRE ATTACK, NVD, CWE, NIST.
Program Analysis	Static analysis.
Programming languages	C, C++, PHP, Python, JavaScript, Matlab.
Databases	MySQL, PostgreSQL.

Research Tools

Static Analysis Tools	Bandit (Python), Flawfinder (C/C++), CodeQL, SonarQube, Semgrep.
Code Review Platforms	GitHub, GitLab, Gerrit.
Repository Mining	GHTorrent, GitHub API.

Research Tools (continued)

AI-based Code Analysis — CodeBERT, Code Llama, ChatGPT, Claude.

Miscellaneous

Scholarly Contribution

2019 — **Reviewer** for DUET journal.

Teaching

2012 – 2020 — **Undergraduate Courses:** Discrete Mathematics, Data Structure, Numerical Methods, Microprocessor & Assembly Language, Digital System Design, Neural Network & Fuzzy system, Algorithm Design & Analysis, Compiler Design, Simulation & Modeling, and Artificial Intelligence.

2021 — **Teaching Assistant:** Python

Awards and Scholarships

2018 — **University Gold Medal** , Bangladesh

2010 — **Prime Minister Gold Medal** , Bangladesh

2009-2010 — **Dean's Award** , Bangladesh

2007-2010 — **University Merit Scholarship** , Bangladesh

2007 — **Ranked first** in University Admission Test

Mentorship

2014 — **Undergraduate students:** Shoishob Roy, Bibhuti vushan, Pankaj mallik

2015 — **Undergraduate students:** Md. Rashed Hossain, Farjana, Md. Shamim Hossain.

2016 — **Undergraduate students:** Shahbuddin Ahmed, Javed rahman , Rocky

Professional Membership

2014-2020 — **Member, Institution of Engineers (IEB),** Bangladesh

2025 — **Member, IEEE**

Member, ACM

References

1. **Dr. Jeffrey Carver**
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   Associate Department Head for Graduate Studies,
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2. **Dr. Md Rayhanur Rahman**
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   University of Alabama,
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